

Recap. In a second-price auction, bidding your true value is a dominant strategy: it maximizes your utility against every profile of other bids. We now ask how to allocate several advertising slots and set payments that retain this guarantee.

Position Auctions

There are k advertising positions (slots) and $n > k$ advertisers (bidders). Each advertiser can receive at most one slot, and each slot can hold at most one advertiser. Slot j has click-through rate (CTR) α_j , the probability of a click, where $1 \geq \alpha_1 \geq \dots \geq \alpha_k > 0$ and $\alpha_j = 0$ for $j > k$. Bidder i has value v_i per click. In slot j at price p per click, their expected utility is

$$u_i = \underbrace{\alpha_j}_{\text{CTR utility per click}} \underbrace{(v_i - p)}_{\text{}} .$$

Values and bids are nonnegative, and ties are broken by a fixed rule. CTRs depend only on the slot. This is the position-auction model [Edelman–Ostrovsky–Schwarz '07; Varian '07].

What is welfare?

Welfare is the sum of all utilities, including the seller's. If bidder i_j receives slot j ,

$$W = \sum_i u_i + u_{\text{seller}} = \sum_{j=1}^k \alpha_j v_{i_j} .$$

Assuming the seller has zero value and cost for the slots, their utility is revenue; payments cancel between bidders and seller.

Claim 1. Assigning slots in decreasing order of value maximizes welfare.

Proof. Regard unallocated bidders as occupying zero-click slots. Take any assignment in which two bidders are out of order: the bidder with value v holds slot j , the bidder with value v' holds slot ℓ , with $j < \ell$ (so $\alpha_j \geq \alpha_\ell$) but $v < v'$. Swapping them changes welfare by

$$(\alpha_j v' + \alpha_\ell v) - (\alpha_j v + \alpha_\ell v') = (\alpha_j - \alpha_\ell)(v' - v) \geq 0 .$$

So swapping never hurts, and repeatedly swapping out-of-order pairs sorts the assignment. Hence some sorted assignment is optimal. $\square$

The Mechanism. Rank bids $b_{(1)} \geq \dots \geq b_{(n)}$ and assign slots in that order. Charge each winner the loss in other bidders' total reported value caused by their presence. To calculate it, remove them and let the bidders below them move up one slot.

Derive the expected payment P_j for the bidder in slot j , and their price per click $p_j = P_j/\alpha_j$.

The gain to bidder (ℓ) below slot j is $(\alpha_{\ell-1} - \alpha_\ell)b_{(\ell)}$. Therefore

$$P_j = \sum_{\ell=j+1}^n (\alpha_{\ell-1} - \alpha_\ell)b_{(\ell)}, \quad p_j = \frac{P_j}{\alpha_j}.$$

The terms above j cancel when subtracting others' welfare with the bidder from others' welfare without them.

Why is $p_j \leq b_{(j+1)}$? What happens with one slot?

The weights $(\alpha_{\ell-1} - \alpha_\ell)/\alpha_j$ are nonnegative and sum to one by telescoping, since $\alpha_n = 0$. Thus p_j is an average of bids at most $b_{(j+1)}$. With one slot, $p_1 = b_{(2)}$.

One advertiser's externality

Let $\alpha = (1, 0.5, 0.2, 0)$ and $v = b = (10, 8, 4, 2)$. For bidder 1, compute the other bidders' total expected value with bidder 1 present and their maximum total expected value without bidder 1. What is their expected payment?

With bidder 1, the others receive expected value $0.5 \cdot 8 + 0.2 \cdot 4 = 4.8$. Without bidder 1, the others move up and receive $1 \cdot 8 + 0.5 \cdot 4 + 0.2 \cdot 2 = 10.4$. Their externality is $10.4 - 4.8 = 5.6$, also equal to $0.5 \cdot 8 + 0.3 \cdot 4 + 0.2 \cdot 2$ from the payment formula.

VCG Beyond Position Auctions

Position auctions are single-parameter: one value per click determines a bidder's value for every slot. With different items, a bidder may need several numbers to describe their values. Such preferences are *multidimensional*; the single-parameter version of Myerson's lemma does not directly apply.

Let Ω be a fixed finite set of feasible outcomes. Bidder i has private value $v_i(\omega)$ for outcome ω , reports $b_i(\omega)$, and has utility $v_i(\omega) - p_i$ when charged total payment p_i . An outcome specifies the whole allocation.

The Vickrey–Clarke–Groves (VCG) mechanism chooses an outcome maximizing reported welfare (breaking ties by a fixed rule):

$$\omega^* \in \operatorname{argmax}_{\omega \in \Omega} \sum_h b_h(\omega).$$

Let W_{-i}^* be the others' maximum reported value when i 's value is omitted. Charge i the difference between this benchmark and others' reported value in the chosen outcome:

$$W_{-i}^* = \max_{\omega \in \Omega} \sum_{h \neq i} b_h(\omega), \quad p_i = W_{-i}^* - \sum_{h \neq i} b_h(\omega^*).$$

In our allocation examples, W_{-i}^* is others' optimal welfare without i : items may remain unallocated and bidders value only their own bundles.

Theorem 1. *For private values and quasilinear utilities, VCG with exact reported-welfare maximization over a fixed finite outcome set is DSIC. Under truthful reports, it maximizes welfare.*

Proof. Fix bidder i 's true valuation and the other reports. If their report selects outcome ω , then

$$\begin{aligned} u_i &= v_i(\omega) - \left[W_{-i}^* - \sum_{h \neq i} b_h(\omega) \right] \\ &= v_i(\omega) + \sum_{h \neq i} b_h(\omega) - W_{-i}^*. \end{aligned}$$

The last term is independent of i 's report. Reporting $b_i = v_i$ makes the allocation rule maximize the first two terms: their true value plus others' reported values. Any other report selects a feasible outcome with no larger value of this expression, so it cannot improve their utility. If all bidders report truthfully, the allocation objective is true welfare. $\square$

Back to slots. Why does this prove truthfulness for the position auction? An outcome is a slot assignment, and each bidder's value is their value per click times their slot's CTR. Sorting maximizes reported welfare, and bidder i 's expected payment p_i in slot j is exactly the externality P_j we derived. The same utility argument applies.

Generalized Second Price

GSP also ranks bids, but charges the bidder in slot j the next bid $b_{(j+1)}$ per click. With one slot this is second price. With multiple slots, truthful bidding need not be dominant [Edelman–Ostrovsky–Schwarz '07; Varian '07].

Exercise. There are two slots, $\alpha = (1, 0.995)$, and three bidders with $v = (10, 4, 2)$. Fix $b_2 = 4, b_3 = 2$. Compare bidder 1's utility from $b_1 = 10$ and $b_1 = 3$ under GSP. Repeat under VCG.

GSP: truthful bidding gives $1(10 - 4) = 6$; bidding 3 gives $0.995(10 - 2) = 7.96$. This beneficial deviation disproves DSIC.

VCG: the top price is $[0.005 \cdot 4 + 0.995 \cdot 2]/1 = 2.01$, giving expected utility 7.99. Bidding 3 gives slot 2 at price 2 and expected utility 7.96, so this deviation does not help.

Does failure of DSIC imply that a game has no Nash equilibrium?

No. DSIC requires truthful bidding to be optimal against every opposing profile. Nash requires each chosen bid to be optimal against the particular opposing bids at the equilibrium.

Acknowledgements

This lecture was developed in part using materials by Tim Roughgarden, and in particular, his book “Twenty Lectures on Algorithmic Game Theory” [Roughgarden, 2016].

References

Tim Roughgarden. *Twenty Lectures on Algorithmic Game Theory*. Cambridge University Press, 2016.